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TECHNICAL PAPER

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Long-Term Field Characterization of Tapered Element Oscillating Microbalance and Modified Tapered Element Oscillating Microbalance Samplers in Urban and Rural New York State Locations

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ABSTRACT

Long-term field comparisons of continuous and integrated filter measurements of mass concentrations of particulate matter (PM) with an aerodynamic diameter less than or equal to 2.5 μm ($PM_{2.5}$) were performed at rural and urban sites in New York State. Two versions of the continuous tapered element oscillating microbalance (TEOM) mass monitor are deployed at each site, in addition to Federal Reference Method filter samplers. Data are grouped into monthly averages to retain and demonstrate seasonal differences. Strong seasonal dependence is observed—the TEOM monitors with the heated sensors are biased systematically low with respect to the Federal Reference Method measurements during the cold season. For the rural site, the average bias for the sample equilibration system (SES)-equipped and standard TEOM monitors is 14 and 24%, respectively. At this location, the TEOM monitor measurements were biased low for all 34 months. For the urban site, the average bias for the SES and standard TEOM monitors is 8 and 18%, respectively. At this location, the TEOM monitor measurements are as likely to be biased high as low during the warm-season months. The hour averaged data from the two versions of the TEOM monitor are also compared, and also indicate that the SES-equipped version of the TEOM monitor captures 7–11% more $PM_{2.5}$ mass at these locations.

IMPLICATIONS

The regulation of $PM_{2.5}$ has led to a flurry of development and testing of PM measuring instrumentation. While the U.S. Environmental Protection Agency has based the current standard on discrete 24-hr averaged filter measurements, most agree that, in the future, continuous mass concentration monitors will be an option for compliance purposes. This paper describes a multi-year program of evaluation of a widely used continuous mass monitor and an evolutionary modification to that monitor. The bias of the continuous monitor with respect to the filter method is examined in detail, as is the seasonal dependence of the bias. Rural and urban differences in the bias estimates are also presented.

INTRODUCTION

The National Ambient Air Quality Standards (NAAQS) for particulate matter (PM) (both PM with an aerodynamic diameter less than or equal to 10 μm [PM_{10}] and PM with an aerodynamic diameter less than or equal to 2.5 μm [$PM_{2.5}$]) are expressed in terms of mass concentrations in micrograms per cubic meter of air. While there is tremendous interest in measuring chemically speciated concentrations of PM,[1–5] bulk mass concentration clearly will be the regulated quantity for the immediate future. With this in mind, it is critically important to establish the quality of data collected under current promulgated $PM_{2.5}$ mass measurement techniques, to identify any significant sources of error associated with the techniques, as well as to evaluate newer technologies that are capable of measuring PM continuously and in real time.

As a result of the attention leading up to and following the 1997 establishment of the NAAQS for fine PM,[6] the air quality measurements community has redoubled its efforts to develop accurate, robust, and easy-to-maintain

instrumentation for measuring $PM_{2.5}$. Airborne PM (or aerosol) is a major issue for air pollution/air quality, visibility, and climate change/global warming. Scientists in each of these disciplines are actively seeking high-quality, long-term PM data sets for studies, assessments, and model evaluations. Air pollution scientists, health effects research scientists, regulators, and policy-makers all point to the need for improved and more detailed data to elucidate scientific and health effects issues related to PM pollution.

Increases in airborne fine PM have been shown in a number of studies to lead to increases in morbidity and mortality, especially for people with respiratory or cardiovascular disease.[7] A number of studies also have shown that small, short-term increases in PM levels were associated with increases in daily mortality and symptoms of certain illnesses.[8–10] These health effects from both short-term and long-term exposure to PM are much smaller than those from smoking tobacco, for example, but are nonetheless significant because of the widespread exposure of large populations to PM pollution.

The existing national PM mass concentration measurement network is neither continuous nor real-time.[11] U.S. Environmental Protection Agency (EPA) has developed a measurement network based on Federal Reference Method (FRM) samplers, which collect one 24-hr integrated filter sample every 3 days.[11] The samplers are run by the states, tribes, or local agencies, and collected samples are often shipped to contract labs for gravimetric analysis; data are typically available 6–8 weeks after sample collection. The network is extensive in scope and has yielded a wealth of useful data, but as noted, this type of data does not fully meet the needs of either air pollution scientists or the health effects community.

Thus, continuous, real-time, robust, and accurate measurement of PM is a major goal of scientists and state and national regulatory agencies. In addition to the scientific reasons discussed, there are at least two additional practical reasons why many states favor continuous instrumentation over the present FRM network. First, continuous measurements are likely to be less expensive because costs associated with extensive shipping and contract laboratory analysis can be avoided. Second, continuous instrumentation would place the complete measurement process under local control, unlike the current system where those states that use contract laboratories are responsible for the measurements but do not have control of the samples and data at all times.

The well-known problems inherent in the measurement of ambient PM include evaporation (and condensation) of semi-volatile compounds from (or onto) the collection medium and the condensation and adsorption of reactive gases from the atmosphere. Recent work quantifying the sources of artifact errors for the FRM[12,13] showed significant evaporative losses from FRM filters. Similar losses are observed for the tapered element oscillating microbalance (TEOM) mass monitor, where evaporation of material from the filter head occurs at the default sensor temperature setting of 50 °C. The SES (sample equilibration system) adaptation for the TEOM mass monitor was developed to reduce the transient measurement errors reported by the TEOM instrument because of water vapor, the most abundant condensable gas in the atmosphere, and to allow instrument operation at temperatures lower than 50 °C.

In this paper, long-term (more than 1 year) evaluations and comparisons of data collected from TEOM and SES-equipped TEOM continuous samplers and FRM samplers at a rural and an urban location in New York State are presented. The recognition and study of seasonal variations, and indeed year-to-year variations, make longer-term data sets and the conclusions drawn from such data more robust and reliable. One of the major goals of the EPA PM Supersite program (PMTACS-NY) is the evaluation of new measurement technologies, specifically with an eye toward technology transfer of robust and easily maintainable instrumentation. Previous studies have compared and evaluated the TEOM monitor and other continuous measurement techniques.[14–17] This work complements and extends the previous work by exploring long-term (multi-year) comparisons at both rural and urban sites in the northeast United States.

SITE INFORMATION

Addison, NY

The Addison site is located at Pinnacle State Park in southwestern New York. The trailer is at a height of ~504 m above sea level (ASL) on the eastern slope of Orr Hill, part of the Allegheny Plateau of southern New York State and northern Pennsylvania. The terrain is typical of the New York/Pennsylvania Twin Tiers region. The site latitude is 42.09° N and its longitude is 77.21° W. The closest trees are ~30 m from the site and consist of a mixture of coniferous and deciduous trees. The surrounding areas include a nine-hole golf course, a picnic area, undeveloped state forestlands, abandoned pastures, a 20-hectare pond, and a handful of residences. The village of Addison (pop. ~1800) is ~4 km to the northwest, and the town of Corning (pop. ~12,000) is 15 km to the northeast. The average annual temperature at this site is 7.8 °C, and the average temperature range is from −18 to 31 °C.

The instrumentation is housed in and on the roof of a 2.4 m × 7.3 m Ekto air monitoring shelter. The shelter is climate-controlled and adjacent to a folding 10-m

meteorological tower. Inlets for all PM instruments are at the same height, 5 m above ground level and 2 m above the shelter roof. The two TEOM monitors are located in the shelter, with 1.8-m long, 1.1-cm ID stainless-steel inlet lines extending straight up through the roof of the shelter. The FRM sampler is a stand-alone unit attached to the roof deck. The FRM sampler was installed at the Addison site in August 1999, and daily FRM samples have been collected since. The standard TEOM monitor was installed in October 1999, and the SES-equipped TEOM monitor was installed in March 2000.

Queens, New York City

New York City Public School 219 (PS 219) is located between the Flushing and Forest Hills neighborhoods in the borough of Queens. The school is directly adjacent to the southeast corner of the Queens College, City University of New York campus. The site latitude is 40.74° N and its longitude is 73.82° W; the altitude is ~25 m ASL. The surrounding neighborhood is predominately urban, and the site is less than 1 km south of the Long Island Expressway (traffic density ~130,000 vehicles per day) and ~1 km east of the Van Wyck Expressway (traffic density ~88,000 vehicles per day), two of the busiest highways in New York City. The site is bordered on the east by a ~1 km^2 cemetery and is located <2 km from two large urban parks. The average annual temperature at this site is 12.8 °C, and the average temperature range is from −9 to 36 °C.

The TEOM monitors and the FRM sampler at this site are situated on the roof of PS219. The school building is two stories high, and the sampling height is ~10 m above ground level. The TEOM monitors are housed in separate temperature-controlled outdoor enclosures, and the FRM sampler is the same stand-alone unit as the one deployed in Addison. The FRM sampler and the standard TEOM instrument were installed at the Queens site in January 2001, and the SES-equipped TEOM instrument was installed in February 2001. FRM samples were collected every third day from late January until mid-May of 2001, at which point daily sampling started at this site.

MEASUREMENT METHODS

24-hr Integrated Filter Samples Using an FRM Sampler

The FRM sampler has been defined, along with the Ambient Air Quality Surveillance Program and the Quality Assurance Program, in a series of Federal Register publications.[6,11,18] Papers by Noble et al.[19] and Peters et al.[20] give good descriptions of the method and the field performance of various FRM samplers.

The FRM sampler used for this study was the Rupprecht & Patashnick Model 2025 Partisol-Plus sequential air sampler. The flow system consists of an inlet with a PM_{10} separator followed by a WINS (Well Impactor Ninety-Six) $PM_{2.5}$ separator,[20] collection on a filter, a flow sensor, and a pump. Flow (16.7 L/min), temperature, and atmospheric pressure sensors associated with the sampler are audited by the site operators monthly and, in New York, by qualified personnel quarterly.

Polytetrafluoroethylene Teflon filters, purchased and quality assured by EPA, are shipped by Research Triangle Institute (RTI) [the New York State Department of Environmental Conservation (NYSDEC) contractor] to the sampling sites as needed. RTI receives the samples collected at the sites, checks to make sure they have remained cold, and places them in an environmentally controlled space where they equilibrate for at least 24 hr before the final weighing operation. All RTI laboratory activity is consistent with EPA guidance. The data are verified by the NYSDEC after examination and comparison of the laboratory results, field data sheets, electronic data downloaded from the samplers, and field-operator notes. The detection limit and precision of the FRM measurements are reported to be 2 and 0.3 $\mu g/m^3$, respectively.[6] One important point to note here is that, while the FRM program collects and analyzes field blank filters, the FRM data are not blank-corrected.

TEOM Mass Monitor

The TEOM mass monitor is based upon a TEOM that is "forced" with an electrical pulse to maintain the amplitude of oscillation without affecting the frequency. A filter module is placed onto the hollow tapered element, and a flow of ambient air is drawn though the filter/tapered element system. As particles collect on the filter, the mass of the filter/tapered element system increases, and the oscillation frequency of the system decreases. This provides the basis for the mass concentration measurement by the TEOM sensor. The commercial TEOM instrument has been described by Patashnick and Rupprecht.[21] The inertial microbalance method (and the TEOM monitor specifically) also has been described in *Methods of Air Sampling and Analysis: Fourth Edition*.[22] Discussed later are aspects of the TEOM monitor of particular relevance to this study.

Flow is drawn into the instrument through a size-selective inlet. The TEOM instruments in this study used the URG-2000–30EH cyclone separator, which is rated for a 2.5-μm size cut at a flow of 16.7 L/min. The size-selected flow is then split by an isokinetic flow splitter into a 3 L/min sample flow and a 13.7 L/min bypass flow. The sample flow is then directed into a ~1-m long,

temperature-controlled, 1.27-cm O.D. inlet flow tube that is maintained at 50 °C for the standard TEOM monitor. The filter and sensor for this unit are also temperature-stabilized at 50 °C. The temperature-controlled sample flow passes through the exchangeable filter cartridge mounted on the TEOM, then through a mass flow controller, and finally exhausts to a piston style vacuum pump. The TEOM monitor operating software allows many variations for controlling and reporting flow rates and flow volumes. (The confusion engendered by the PM_{10} standard being referenced to standard flows and $PM_{2.5}$ standards being referenced to local volumetric flows practically requires such flexibility.) For comparison with FRM measurements, all TEOM monitor flow volumes in this study are referenced to local volumetric flows. Weekly check sheets for the TEOM monitors were completed by site operators and sent to the NYSDEC program manager. TEOM monitor flows and temperatures were audited quarterly by NYSDEC personnel and annually by NYSDEC quality assurance personnel. Precision of the TEOM monitor for hour averaged data has been determined from collocated measurements to be ± 1.5 $\mu g/m^3$.[22] Accuracy is more difficult to assess, because the nature of the aerosol itself will affect the ability of the instrument to sample and detect the aerosol. Under the highly favorable conditions of a stable, nonvolatile aerosol in the absence of condensable or reactive gases, the microbalance could exhibit a lower quantifiable limit as low as 0.3 $\mu g/m^3$. In practice, most users of the TEOM monitor use a lower quantifiable limit of 2 $\mu g/m^3$ or higher for hour averaged data.

The TEOM monitor operating software has a 2-sec sampling period for individual measurements of oscillation frequency. The updated frequencies lead to updated total mass data, which are folded into the computation of the 10-min smoothed total mass and mass concentration. Averaging times are user-selectable, but the default settings give smoothed averages for total mass and mass concentration with a rise time of 10 min for 90% response to a step change in concentration.

While the TEOM monitor allows internal scaling of the data, and the instrument has default scaling factors that multiply the measured values by 1.03 and add an offset of 3.0 $\mu g/m^3$, the data used in this study are unscaled. At both sites, minute and hour averaged data are collected by a data logger. At Addison, the minute data are downloaded to an on site computer and used as the base averaging interval. At Queens, only the hour averaged data computed by the data logger from the minute averaged data are downloaded via modem to a central computer. The Addison data were verified and flagged at Atmospheric Sciences Research Center, and the Queens data was verified and flagged by NYSDEC. The Addison minute data were averaged to hour data with the requirement that there be 45 valid minute data points to produce a valid hour average. Addison and Queens hour average data were further averaged to 24-hr averages with the requirement that there be 21 valid hour averages to produce a valid 24-hr average.

SES-Equipped TEOM Mass Monitor

The standard TEOM mass monitor configuration utilizes a stabilization temperature of 50 °C, primarily to mitigate the large swings in mass measured by the instrument caused by the condensation and evaporation of water vapor at lower operating temperatures. The interfering signals caused by changes in water vapor during a front passage can easily overwhelm the signal caused by atmospheric particles. The operating temperature of 50 °C helps to hold down this interference. Unfortunately, the 50 °C operating temperature also leads to the evaporation of many semi-volatile species like ammonium nitrate (NH_4NO_3) and numerous organic compounds. Given the significant issues surrounding semi-volatile PM, and the desire to improve the TEOM instrument to more closely measure the "true" atmospheric aerosol, the SES modification to the TEOM mass monitor was developed to allow conditioning of the PM sample stream to a lower humidity and temperature level.[23]

The SES system is distinguished by two changes from the standard TEOM monitor: the sample flow passes through a Nafion dryer bundle before being introduced into the sensor unit, and the sensor unit is maintained at a temperature lower than 50 °C—for this study, the sensor temperature was set to 30 °C. As described by Meyer et al.,[23] the SES dryer for the sample stream uses a bundle of 18 tubes of Nafion that has been designed to minimize the potential for particle loss at the entry point. As shown by the SES flow in Figure 1, a pair of dryers is used to obtain better efficiency of humidity reduction. Low-pressure exhaust air on the downstream side of the mass flow controllers is used as the purge flow for the dryer bundles. The lower sensor temperature is the major advantage of the SES-equipped TEOM monitor over the standard configuration. One expects significantly better collection efficiency for semi-volatile species that have high vapor pressures in the 30–50 °C temperature range. Any fixed temperature for the sensor head introduces some problems, however. If the sensor is held constant at 30 °C and the ambient temperature goes above 30 °C, there is the potential for condensation onto the filter. More significantly, even at 30 °C, the sensor head is quite

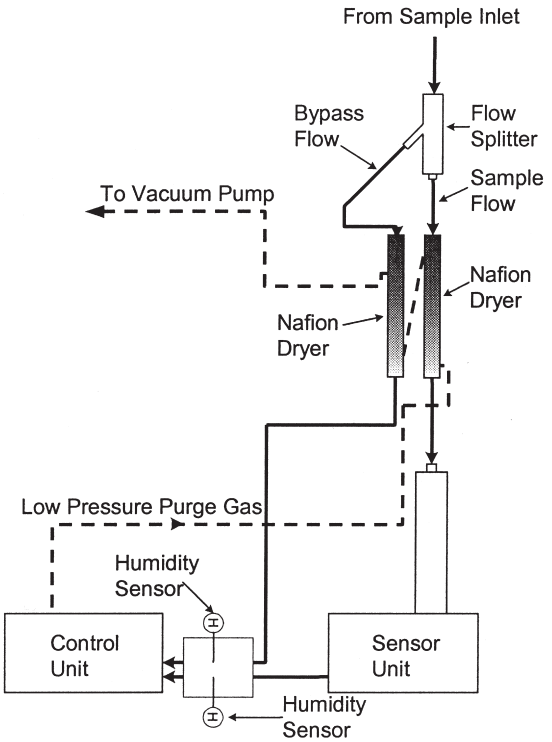


Figure 1. Flow schematic of the SES-equipped TEOM mass monitor.

a bit warmer than ambient temperature during the winter months at the two test sites, and many semi-volatile compounds will evaporate upon warming or after they have been collected onto the filter.

RESULTS

Analysis Methodology

The FRM filter data were verified and flagged by the weighing laboratory and NYSDEC. The 24-hr integrated average mass concentrations used here were also reported into the Air Quality System (AQS) database. The 24-hr TEOM monitor average mass concentrations were calculated as described. Data comparisons are done pair-wise—standard TEOM monitor versus FRM, SES-equipped TEOM monitor versus FRM, and SES-equipped versus standard TEOM monitor. As detailed in the Measurement Methods section, the $PM_{2.5}$ size selectors are different for the FRM and the TEOM monitors. This difference may add a small error to the analysis. Only days for which valid FRM, standard TEOM monitor, and SES-equipped TEOM monitor mass concentration data were available are used in this analysis. Linear regressions and correlation coefficients are calculated with the FRM mass concentrations as the independent variable for comparisons between TEOM monitor and FRM data, and the standard TEOM monitor mass concentrations are used as the independent variable for the standard TEOM monitor–SES-equipped TEOM monitor data comparison.

For comparisons between TEOM monitor and FRM data, the slope of the regression line is suggestive of the ratio of $PM_{2.5}$ mass collected by the two measurements or the fraction of FRM mass collected by the TEOM monitor. The intercept of the unconstrained least squares fit regression line is suggestive of the systematic offset between the two measurements. The data in this study are highly correlated and tend to have correlation coefficients (R^2) of 0.9 or higher. Because the data are highly correlated, and because fluctuations in the y intercepts of the regression lines can mask subtle but important changes in the ratio of $PM_{2.5}$ mass collected by the measurements, a second linear least squares regression fit to each data set was performed with the constraint that the regression line passes through (0,0). If the R^2 values remain high for the constrained regressions, these zero-intercept regression slopes give an offset independent measure of the ratio of $PM_{2.5}$ mass collected by the two instruments.

For comparisons of standard and SES TEOM monitor measurements, one can use hour-averaged data. The standard TEOM measurement is chosen as the independent variable because of the large number of deployed standard TEOM monitors and databases derived there from worldwide. In this case, the slope indicates the ratio of mass collected by the SES-equipped monitor to that collected by the standard TEOM monitor. The intercept indicates some combination of possible systematic offsets, and its variability indicates in some measure the precision of the measurement techniques.

Because the characteristics of the aerosol composition change seasonally,[24–26] and because the temperature difference of the TEOM sensor head and the ambient air also changes dramatically with season, comparisons are done by month. That is, TEOM monitor and FRM data are broken out into monthly sets, and the unconstrained and constrained linear regressions are calculated for each month.

Average PM Mass Concentrations

As noted, the FRM mass concentrations are not corrected for field blanks, which introduces a positive bias to these measurements. For the New York State FRM network, the average field blank, expressed as an effective mass concentration by dividing by the average flow volume, is 0.27 $\mu g/m^3$, with a standard deviation of 0.26 $\mu g/m^3$. These values are computed from 892 field blanks measured during 2001 and 2002. Data presented and used here are the FRM data used by EPA, without blank corrections. Figure 2 shows the monthly mean $PM_{2.5}$ mass concentrations measured by the FRM sampler observed at the Addison site. Also shown are error bars, which indicate plus and minus 1 standard deviation of the daily values. Evident

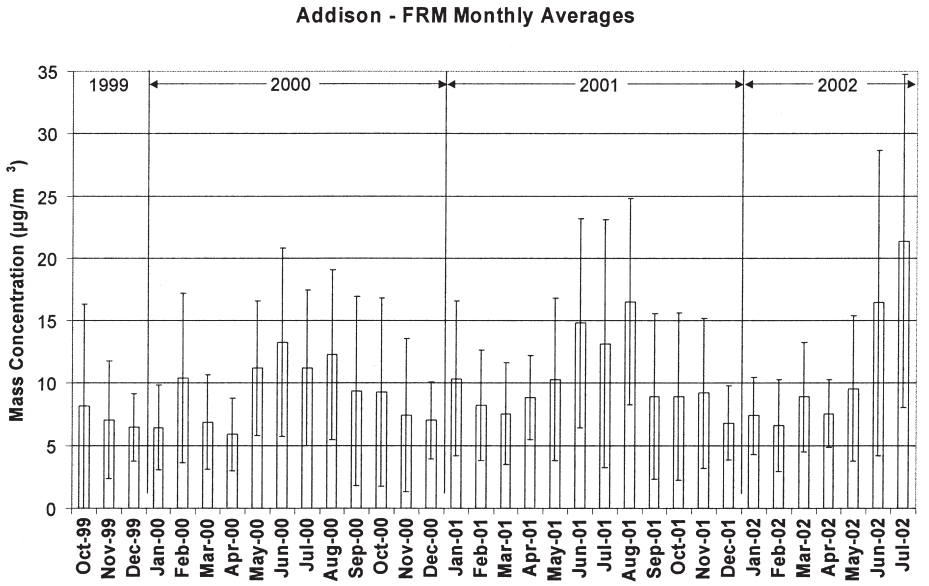


Figure 2. Mean mass concentration by month for valid FRM filter samples at the Addison site. The error bars indicate ±1 SD about the mean.

from the figure are two important points: there is seasonal variation with highest values measured during the summer months, and there is year-to-year variation in the exact timing and magnitude of the peak values. While the winter periods show generally lower measured mass concentrations, February 2000 and January 2001 measurements indicate possible secondary maxima.

Figure 3 shows the monthly mean $PM_{2.5}$ mass concentrations measured by the FRM sampler at the Queens site. Again, the highest values occur in the summer months, but notice that July 2001 was, on average, extremely clean, and the mean for that month is below the values for February 2001 and January 2002. It would seem that more data are required to clearly establish the magnitude of the seasonal variation at this site and to confirm or reject the existence of a second seasonal maximum during the cold winter months. During the summer months, the measured mass concentrations at Addison and Queens are relatively close, and in June 2002, the Addison monthly average exceeds the Queens monthly

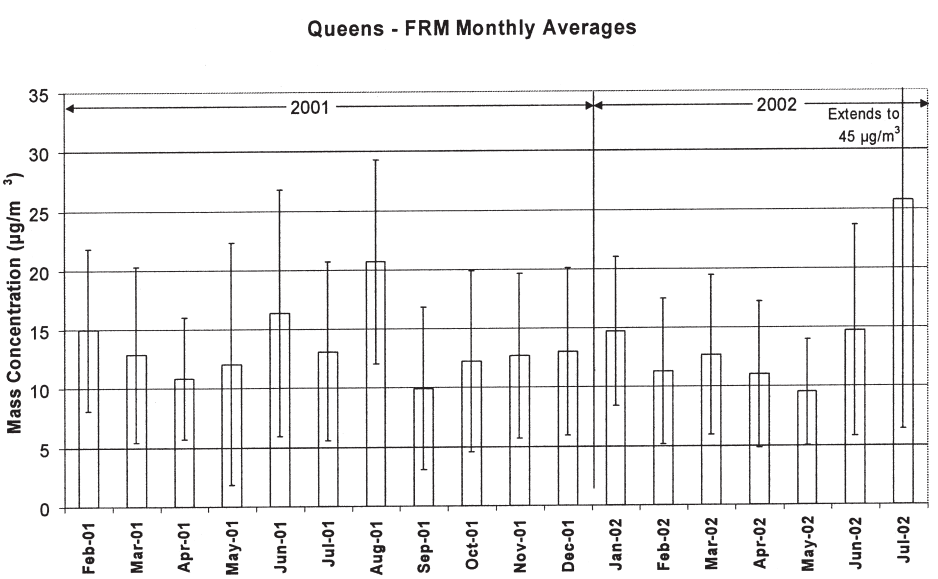


Figure 3. Mean mass concentration by month for valid FRM filter samples at the Queens site. The error bars indicate ±1 SD about the mean.

average because of a large regional haze event that affected Addison more than Queens. This contrasts with the winter periods in which the concentrations in Queens are ~5 $\mu g/m^3$ greater than those in Addison. The summer aerosol is more regional and sulfate (SO_4^{2-})-dominated, and the urban winter aerosol is more local and nitrate (NO_3^-) and carbon-dominated.[27]

TEOM/FRM Comparisons

Figure 4 shows scatter plots of all the 24-hr Addison data used in this study. There are 843 data points plotted in Figure 4a and 715 data points plotted in Figure 4b. The data are highly correlated, as others have seen when comparing TEOM monitor data with data from integrated filter samplers.[23,28,29] Figure 4a shows the standard TEOM monitor data plotted versus the FRM data, and Figure 4b shows the SES-equipped TEOM monitor data plotted versus the FRM data. Two regression lines are shown for each figure; the lighter line calculates the slope and intercept, and the darker line calculates the slope for the line with the intercept constrained to 0. These figures indicate that, on average, the TEOM monitors measure a slightly smaller amount of $PM_{2.5}$ mass than the FRM at this location and that there appears to be a systematic offset or bias as evidenced by the negative intercepts.

To understand the reason for this shortfall, and to gain additional insight into the measurement process itself, it is useful to break up the data into month-long periods. Figure 5 shows scatter plots for a winter and a summer month for each TEOM monitor. Figures 5a and b are scatter plots of the standard and SES-equipped TEOM monitors versus the FRM for July 2002, and Figures 5c and d are scatter plots of the standard and SES-equipped TEOM monitors versus the FRM for December 2001. The slopes are much higher for the July data and are not significantly different than one for the SES-equipped TEOM monitor. The December data are less well correlated and have slopes quite a bit lower than 1. The poorer correlation is attributed to the much larger temperature difference between ambient temperatures (monthly average 0 °C) and TEOM sensor temperatures (50 and 30 °C) than is the case during the summer months. Some, but certainly not all, of the PM collected during the colder months is lost from the heated TEOM sensor.

Slopes and intercepts for all monthly data sets for the Addison site are displayed in Figure 6. Figure 6a displays the fitted slopes for the standard and SES-equipped TEOM monitor data versus the FRM data linear least squares regression, and Figure 6b shows the fitted intercepts. Correlation coefficients for these monthly regressions are quite high, with R^2 typically greater than 0.9 (average R^2 values for the two data sets displayed in Figure 6a are 0.936 and 0.949). The single month (January 2000) with R^2 less than 0.8 is indicated by a vertically striped bar, and the months with R^2 between 0.8 and 0.9 are indicated by horizontally striped bars. From October 1999 to February 2000, the only TEOM data available are from the standard TEOM monitor. The seasonal dependence of the slope values is evident in Figure 6a, and the slopes for the SES-equipped TEOM monitor versus FRM regressions are larger than the slopes of the standard TEOM monitor versus FRM regressions in most (but not all) months. The seasonal variation is assumed to be caused predominately by two factors: (1) the temperature difference between the

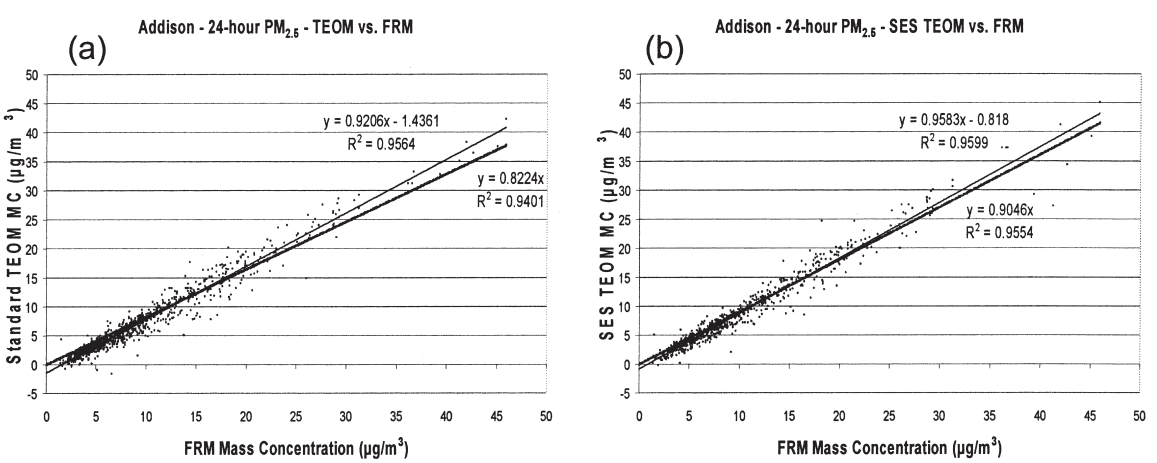


Figure 4. (a) Correlation scatter plot, using all valid pairs of data, of the standard TEOM monitor mass concentrations vs. the FRM mass concentrations at the Addison site. The TEOM monitor data have been averaged to 24-hr values to match the filter samples. The lighter line shows the result of the unconstrained linear regression fit, and the darker line shows the results of the zero-intercept regression fit. (b) Correlation scatter plot, using all valid pairs of data, of the SES-equipped TEOM monitor mass concentrations vs. the FRM mass concentrations at the Addison site. The TEOM monitor data have been averaged to 24-hr values to match the filter samples. Lighter and darker lines are as in Figure 4a.

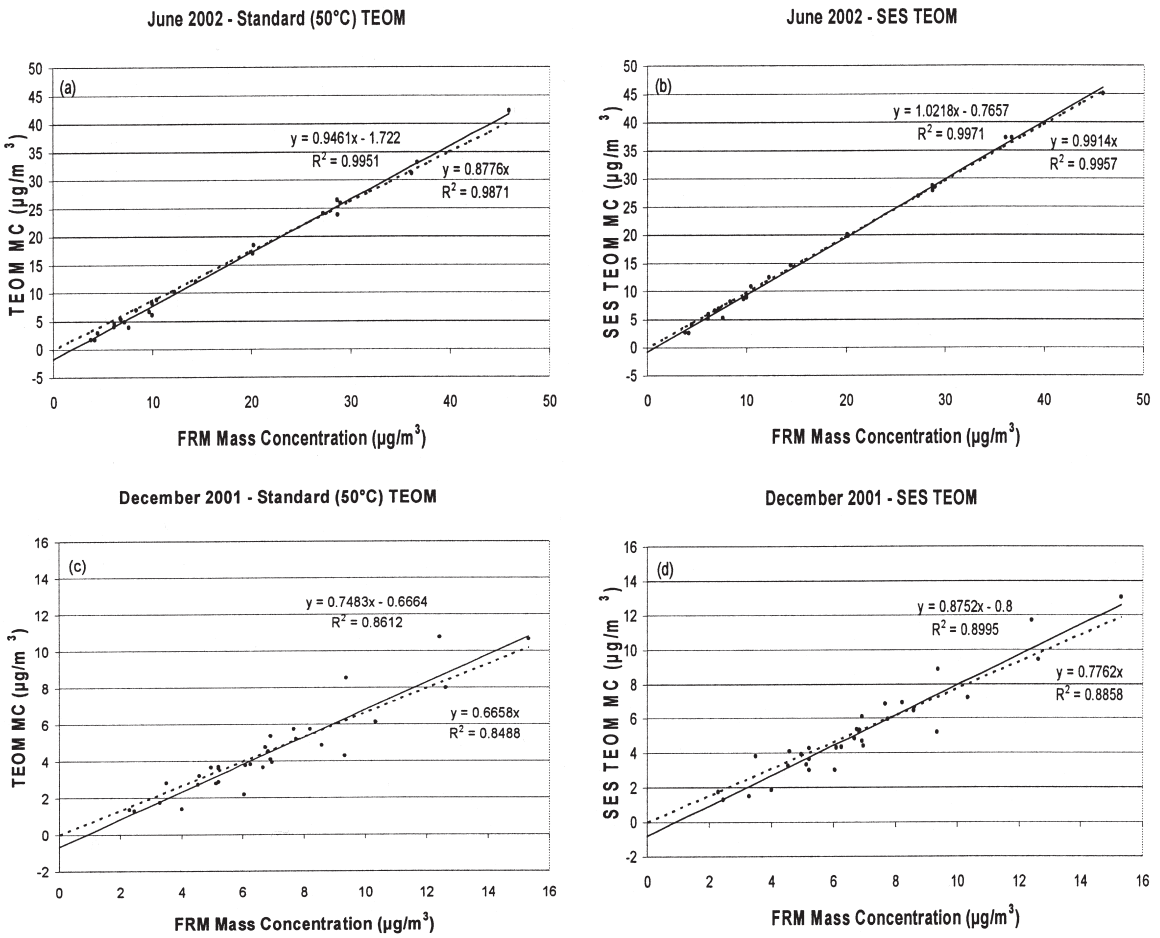


Figure 5. (a) Correlation scatter plot, using July 2002 data, of the standard TEOM monitor mass concentrations vs. the FRM mass concentrations at the Addison site. Solid and dashed lines are as in Figure 4. (b) Correlation scatter plot, using July 2002 data, of the SES-equipped TEOM monitor mass concentrations vs. the FRM mass concentrations at the Addison site. Solid and dashed lines are as in Figure 4. (c) Correlation scatter plot, using December 2001 data, of the standard TEOM monitor mass concentrations vs. the FRM mass concentrations at the Addison site. Solid and dashed lines are as in Figure 4. (d) Correlation scatter plot, using December 2001 data, of the SES-equipped TEOM monitor mass concentrations vs. the FRM mass concentrations at the Addison site. Solid and dashed lines are as in Figure 4.

ambient environment and the TEOM sensor, and (2) the chemical composition and volatility of the PM. The fitted intercepts in Figure 6b are nearly all below 0 and, for a given instrument, can vary month to month by values approaching 3 $\mu g/m^3$. One interpretation of the negative intercepts is that the TEOM monitor results are biased low with respect to the FRM samples. This will be explored further in the section on bias estimates.

The zero-intercept regression slopes for the Addison monthly data are displayed in Figure 7. The correlation coefficients for these fits are also quite high—the average difference of the R^2 values between the unconstrained and zero-intercept regression fits is 1–2%, and the maximum difference in R^2 is 9%. As in Figure 6a, the single month with R^2 less than 0.8 is indicated by a vertically striped bar, and the months with R^2 between 0.8 and 0.9 are indicated by horizontally striped bars. The zero-intercept slopes fall into a much narrower range than do the slopes in Figure 6a. In addition, all of the zero-intercept slopes are less than 1, even though quite a few of the SES-equipped TEOM monitor slopes nearly reach 1. Because the slopes are less than 1 for all months, there is some combination of negative bias for the TEOM monitor and positive bias for the FRM. The small positive bias in the FRM data is known because it is not blank-corrected. The positive FRM bias would increase all slopes by 0.015–0.045 (depending on the month).

Queens standard and SES-equipped TEOM monitor data are plotted versus FRM data in Figure 8. There are 383 valid sets of data used in these plots. As with the Addison data, Figure 8a shows the standard TEOM monitor versus FRM scatter plot and the full regression and zero-intercept regression lines. Figure 8b shows the scatter plot for the SES-equipped TEOM monitor versus the FRM. Monthly scatter plots and linear least squares regression fits also were done for the Queens data. The Queens FRM sampler

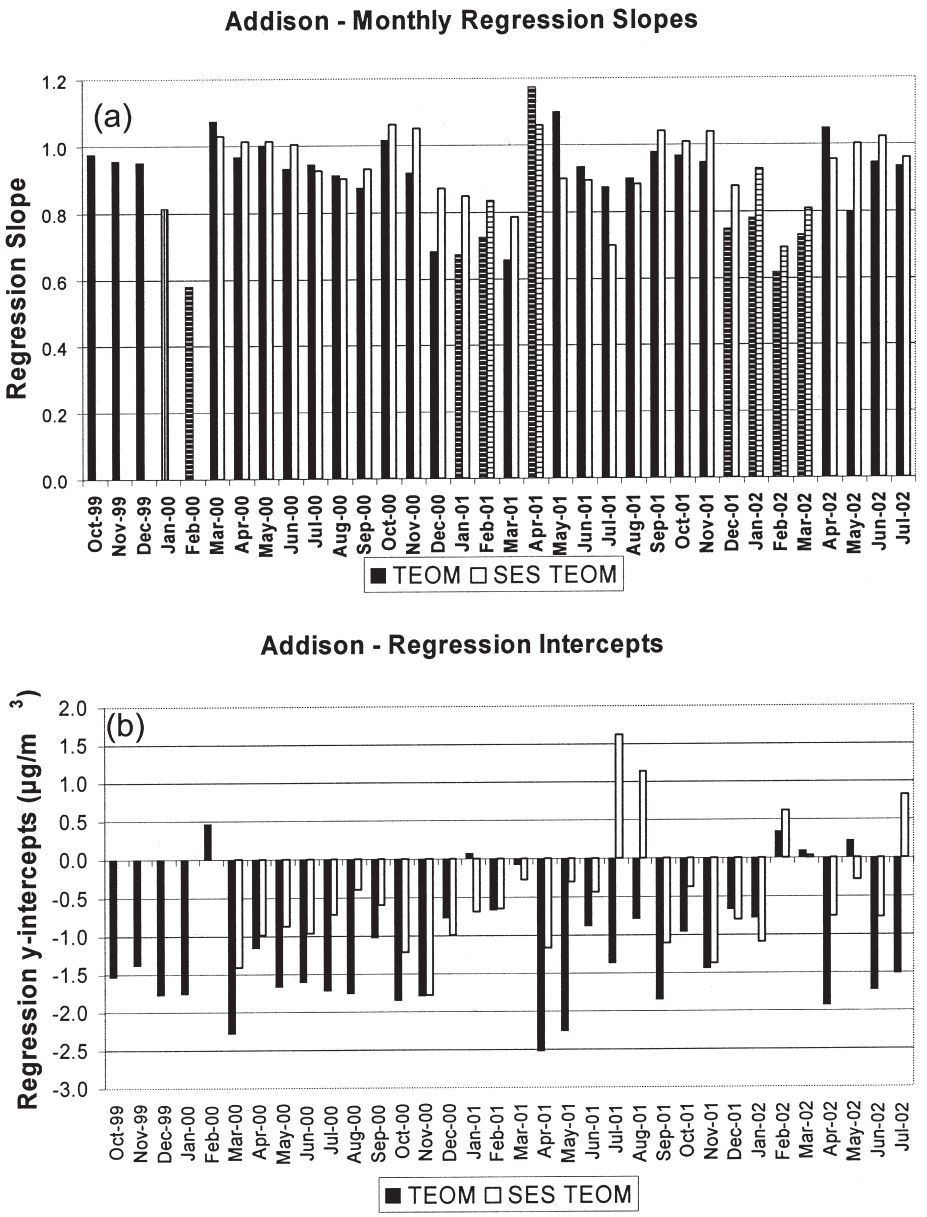


Figure 6. (a) Fitted linear regression slopes for the standard and SES-equipped TEOM monitor mass concentrations vs. the FRM mass concentrations separated by month at Addison. Dark solid bars denote months for which the R^2 values of the standard TEOM monitor vs. FRM regressions are greater than 0.9. Solid white bars indicate months for which the R^2 values of the SES-equipped TEOM monitor vs. FRM regressions are greater than 0.9. Bars with horizontal stripes indicate months for which the R^2 values are between 0.8 and 0.9, and the vertical stripes indicate an R^2 value less than 0.8. (b) Fitted linear regression intercepts for the standard and SES-equipped TEOM monitor mass concentrations vs. the FRM mass concentrations separated by month at Addison. Dark bars indicate the standard TEOM monitor vs. FRM intercepts, and white bars indicate the SES-equipped TEOM monitor vs. FRM intercepts.

was operated on a 1-in-3 days schedule until mid-May 2001, and daily since then, so there are fewer data points for the February–May 2001 monthly data sets.

Slopes and intercepts for all monthly data sets for the Queens site are displayed in Figure 9. Figure 9a displays the fitted slopes for the least squares linear regressions of the standard and SES-equipped TEOM monitors versus the FRM, and Figure 9b shows the fitted intercepts. Correlation coefficients for these monthly regressions are quite high, with R^2 typically greater than 0.9. The average of the R^2 values for the data displayed in Figure 9a are 0.922 and 0.929. Data with correlation coefficients between 0.8 and 0.9 are indicated by bars with horizontal stripes. The seasonal dependence of the slope values is evident in Figure 9a, and the slope for the SES-equipped TEOM monitor versus FRM regression is larger in most (but not all) months. The March 2001 slope, with a value of 0.46, is especially low and appears to be dominated by

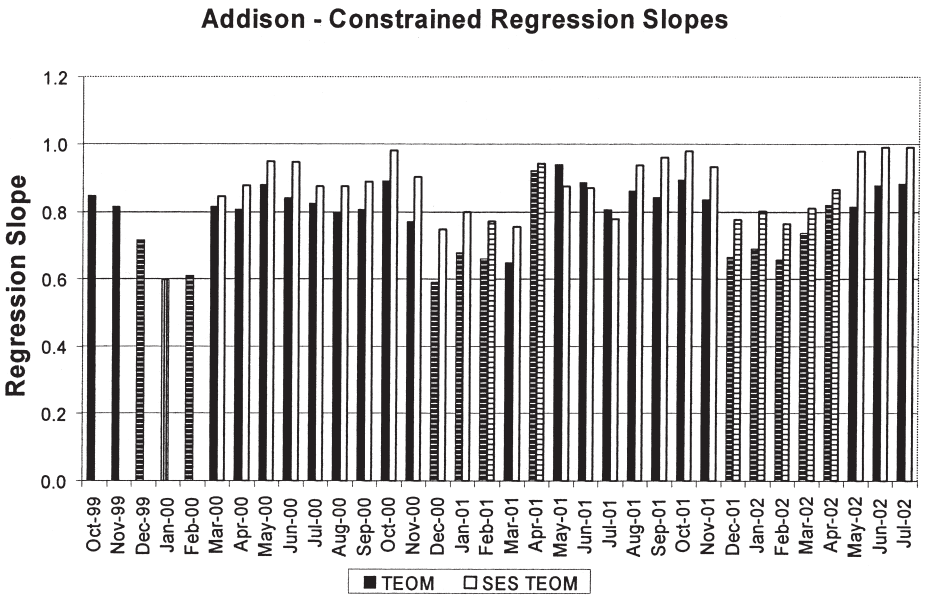


Figure 7. Fitted zero-intercept regression slopes for the standard and SES-equipped TEOM monitor mass concentrations vs. the FRM mass concentrations separated by month at Addison. Dark solid bars denote months for which the R^2 values of the standard TEOM monitor vs. FRM regressions are greater than 0.9. Solid white bars indicate months for which the R^2 values of the SES-equipped TEOM monitor vs. FRM regressions are greater than 0.9. Bars with horizontal stripes indicate months for which the R^2 values are between 0.8 and 0.9, and the vertical stripes indicate an R^2 value less than 0.8.

the two highest of the 10 data points available for that month. The fitted intercepts in Figure 9b exhibit a greater number of quite large positive and negative values than does the Addison site, and it is notable that the largest positive intercept from March 2001 corresponds to the very low slope in Figure 9a.

The zero-intercept regression slopes for the Queens monthly data are displayed in Figure 10. The correlation coefficients for these fits are also quite high, differing significantly from those of the unconstrained fits only when the fitted intercept differs significantly from 0. Excluding the March 2001 data, where the R^2 value for the zero-intercept regression is nearly 14% lower than the R^2 for the unconstrained regression, the average difference in R^2 values is 0.5%. The single month (March 2001) with R^2 less than 0.8 is indicated by a vertically striped bar, and the months with R^2 between 0.8 and 0.9 are indicated by horizontally striped bars. The zero-intercept slopes fall

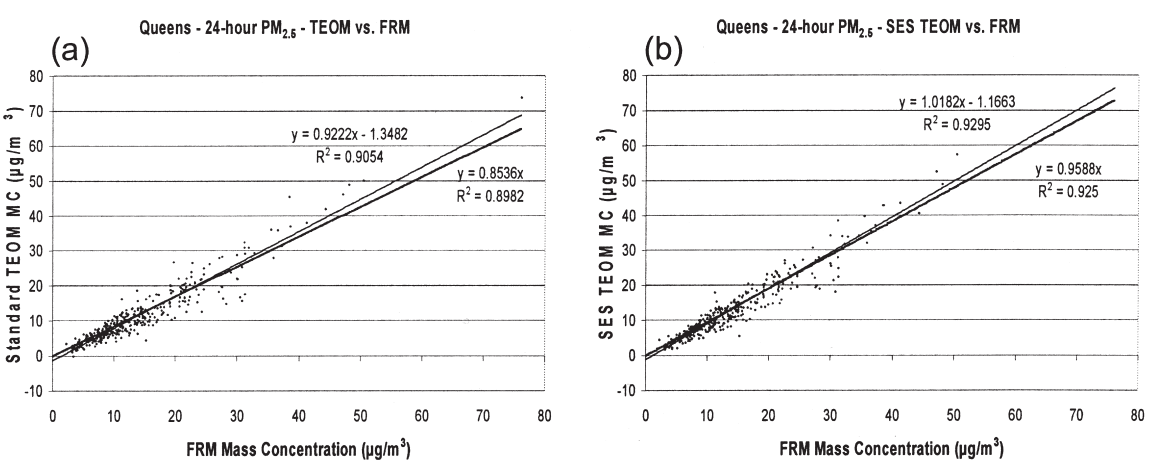


Figure 8. (a) Correlation scatter plot, using all valid pairs of data, of the standard TEOM monitor mass concentrations vs. the FRM mass concentrations at the Queens site. The TEOM monitor data have been averaged to 24-hr values to match the filter samples. The lighter line shows the result of the unconstrained linear regression fit, and the darker line shows the results of the zero-intercept regression fit. (b) Correlation scatter plot, using all valid pairs of data, of the SES-equipped TEOM monitor mass concentrations vs. the FRM mass concentrations at the Queens site. The TEOM monitor data have been averaged to 24-hr values to match the filter samples. Lighter and darker lines are as in Figure 4a.

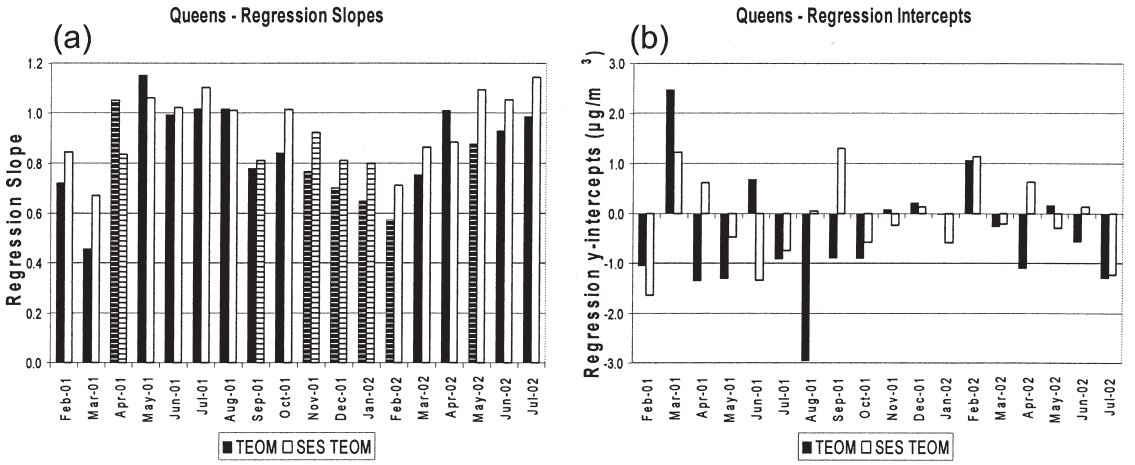


Figure 9. (a) Fitted linear regression slopes for the standard and SES-equipped TEOM monitor mass concentrations vs. the FRM mass concentrations separated by month at Queens. Dark solid bars denote months for which the R^2 values of the standard TEOM monitor vs. FRM regressions are greater than 0.9. Solid white bars indicate months for which the R^2 values of the SES-equipped TEOM monitor vs. FRM regressions are greater than 0.9. Bars with horizontal stripes indicate months for which the R^2 values are between 0.8 and 0.9. (b) Fitted linear regression intercepts for the standard and SES-equipped TEOM monitor mass concentrations vs. the FRM mass concentrations separated by month at Queens. Dark bars indicate the standard TEOM monitor vs. FRM intercepts, and white bars indicate the SES-equipped TEOM monitor vs. FRM intercepts.

into a much narrower range than do the slopes in Figure 9a. Still, the seasonal dependence is evident. The winter-to-summer variation in slopes is greater for the Queens data than for the Addison data, an observation attributed to a more stable PM mixture during the summer in Queens. Chemically speciated measurements could confirm or disprove this assumption. The Queens data do contain a number of months in which the zero-intercept slopes are greater than 1, in most cases only by a few percent. In the months of April, May, and June 2001, the standard TEOM monitor monthly mean mass concentrations are greater than the SES-equipped TEOM monitor monthly means (this is also true for the Addison data). In addition, during May–August of 2001, the monthly means of either the standard or SES-equipped TEOM monitor measurements, or both, are greater than the FRM monthly means. This type of result could be caused by instrument problems with the TEOM monitors. However, because both sets of TEOM monitors, at sites separated by hundreds of kilometers, show the same features, the problem would have to be systematic. The other possibility is an unusual composition of the ambient aerosol for just those few months. This issue is discussed in more detail later.

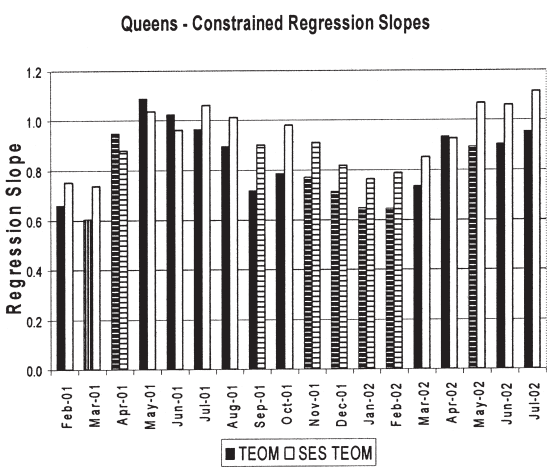


Figure 10. Fitted zero-intercept regression slopes for the standard and SES-equipped TEOM monitor mass concentrations vs. the FRM mass concentrations separated by month at Queens. Dark solid bars denote months for which the R^2 values of the standard TEOM monitor vs. FRM regressions are greater than 0.9. Solid white bars indicate months for which the R^2 values of the SES-equipped TEOM monitor vs. FRM regressions are greater than 0.9. Bars with horizontal stripes indicate months for which the R^2 values are between 0.8 and 0.9, and the vertical stripes indicate an R^2 value less than 0.8.

Bias Estimates

Given the complexity of atmospheric aerosol, and the diversity of techniques for measuring it, estimates of the relative bias of various measurement techniques are quite useful. As noted, the FRM does not constitute a reference standard, nor is it a true measurement of ambient PM. (Note that we have not defined, and will not attempt to provide a rigorous definition for, the term "ambient PM.") Still, the NAAQS for PM is expressed in terms of measurements made using the FRM.[6] This makes FRM measurements the best choice for the measurement technique against which the other techniques will be compared.

The TEOM monitor bias with respect to the collocated FRM measurement is calculated from the monthly least squares regression slopes and intercepts and the monthly average FRM mass concentrations. As noted, the regression intercept gives an indication of the bias. This indication is only a bias estimate if the slope of the regression line is unity, which is not the case in general. A formula that combines the regression intercept with the slope and FRM average mass concentration is as follows:

$$Bias = b + (m - 1)*\overline{MC}_{\mathrm{FRM}}$$

In this equation, b is the regression intercept, m is the regression slope, and MC_{FRM} is the average FRM mass concentration for the month under consideration. As with the previous slopes, bias estimates are calculated by month for both sites. Note that one could easily compute the blank-corrected bias by simply using a blank-corrected FRM mass concentration. One also can calculate a bias estimate from the zero-intercept regression slope. The estimates obtained from the zero-intercept slopes are on the average ~15% lower than those calculated using the formula. Estimates from the unconstrained linear regressions are presented because they retain slightly more information leading into the final calculation.

Figure 11 presents the bias estimates for the standard and SES-equipped TEOM monitors at the Addison site. The bias is negative for all months, indicating that the TEOM monitors collect less mass than the FRM, on average. (If blank-corrected FRM data were used, the July 2002 bias estimate would be positive.) There is a clear seasonality to this bias, showing evidence of greater bias in both winter and summer. It is reasonable to assume the winter bias is caused in large part by volatilization of NH_4NO_3 from the TEOM sensor head, and that the summer bias is caused by volatilization of photochemically produced organic species. For the standard TEOM monitor (black bars), winter month bias estimates sometimes reach 3–4 $\mu g/m^3$, which is ~40% of the FRM monthly average. A bias of 3 $\mu g/m^3$ for a summer month (such as July 2001) is more like 20–25% of the FRM monthly average. The SES-equipped TEOM monitor clearly has a smaller bias. The seasonality is still quite clear, but the largest bias monthly estimates are ~2.5 $\mu g/m^3$, or 25% of the FRM monthly average.

Figure 12 is the same as Figure 11, but for the Queens site. Unlike the Addison site, the bias estimates are positive for some, but not all, of the warmer months. These calculations only give estimates of the monthly averaged bias, and estimates within ± 0.5 $\mu g/m^3$ of 0 are not significantly different from 0. The negative bias in winter months is clearly evident, and for the standard TEOM monitor, this bias estimate reaches over 5 $\mu g/m^3$, or 30–35% of the FRM monthly average mass concentration. The data indicate that the SES-equipped TEOM monitor, in particular, generally collects about the same or even more aerosol mass than does the FRM sampler at this location during the warm season.

The summary of bias estimates for the rural Addison site and the urban Queens site is given in Table 1. These average bias estimates are calculated using the monthly averages displayed in Figures 11 and 12. The methodology gives equal weight to each monthly average and may be slightly different from a calculation using daily averages. The biases are presented both in absolute terms (as $\mu g/m^3$) and in percent of the monthly averaged FRM mass

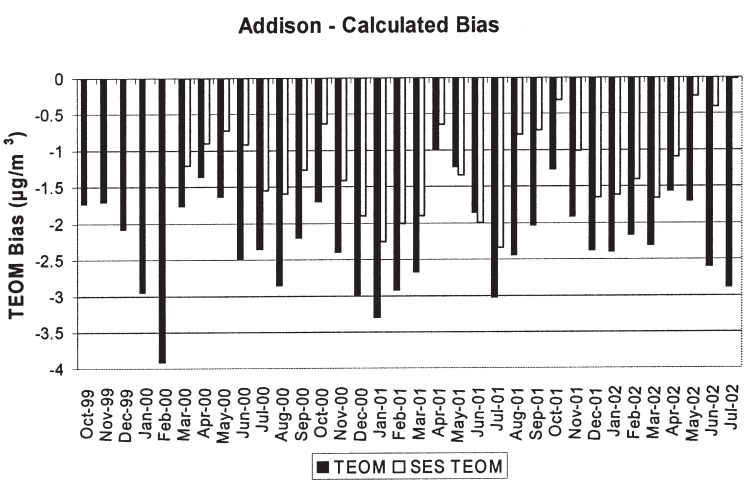


Figure 11. Estimated bias in $\mu g/m^3$ for the standard and SES-equipped TEOM monitors with respect to the FRM filter mass concentrations at the Addison site. Dark bars indicate the estimated bias of the standard TEOM monitor, and white bars indicate the estimated bias of the SES-equipped TEOM monitor.

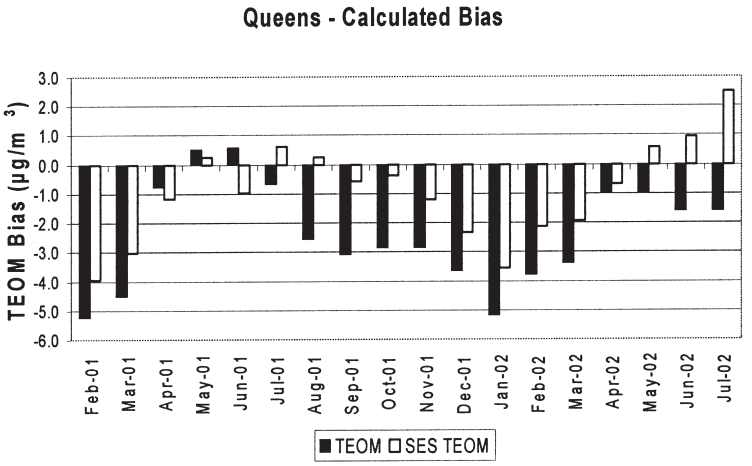


Figure 12. Estimated bias in $\mu g/m^3$ for the standard and SES-equipped TEOM monitors with respect to the FRM filter mass concentrations at the Queens site. Dark bars indicate the estimated bias of the standard TEOM monitor, and white bars indicate the estimated bias of the SES-equipped TEOM monitor.

concentrations. While the absolute values of the urban and rural bias estimates are similar, expressed as a percentage of the FRM, the relative bias for the TEOM monitors in the urban location are 6% lower. The bias estimates also are calculated on an annualized basis, that is, for each contiguous 12-month period during which measurements were made for this study. The annualized bias estimates are quite close to those calculated for all data, differing at most by ~12%. The annualized bias estimates are compared with annualized FRM measurements to determine the relative annualized bias. The different averaging periods for the FRM measurements lead to the very small difference in relative bias, even when the absolute bias differs by 10–12%.

The results presented so far indicate that the SES-equipped TEOM monitor does collect more aerosol mass than the standard TEOM monitor, as expected because of the cooler sensor temperature. The head-to-head comparison of the hour averaged TEOM monitor data sets is now presented as a confirmation of the results, and as a means to further explore the continuous data.

Standard TEOM/SES-Equipped TEOM Data Comparisons

Figure 13 shows the scatter plot of all hourly TEOM monitor data for Addison, with the standard TEOM monitor mass concentration plotted as the independent variable, and SES-equipped TEOM monitor mass concentration plotted as the dependent variable. Also plotted are two linear least squares regression lines, with the equations for the lines and their correlation coefficients displayed. Again, there is very strong correlation, but there is also evidence of outlying points. Most of the outliers are below and to the right of the regression lines. There are two explanations for these points, which indicate a standard TEOM monitor reading significantly larger than the SES-equipped TEOM monitor reading. Many of the clusters of outliers represent data points from mid-April through July 2001, an anomalous period indicated by small open circles on the plot. Removing data from this period gives a much tighter correlation and higher slopes. The second source of outliers tends to be more isolated points, and small groups of points, that are likely caused by positive swings in relative humidity.[30] The standard TEOM monitor (without the Nafion dryer) exhibits large transient

Table 1. Calculated bias estimates for the Addison and Queens sites.

	Standard TEOM Absolute ($\mu g/m^3$)	Standard TEOM % FRM	SES TEOM Absolute ($\mu g/m^3$)	SES TEOM % FRM
Addison–all data	−2.23	−24.5%	−1.22	−13.6
Addison–annualized	−2.24	−22.9%	−1.37	−13.8
Queens–all data	−2.37	−18%	−0.93	−7.6
Queens–annualized	−2.48	−18.5%	−1.03	−7.7

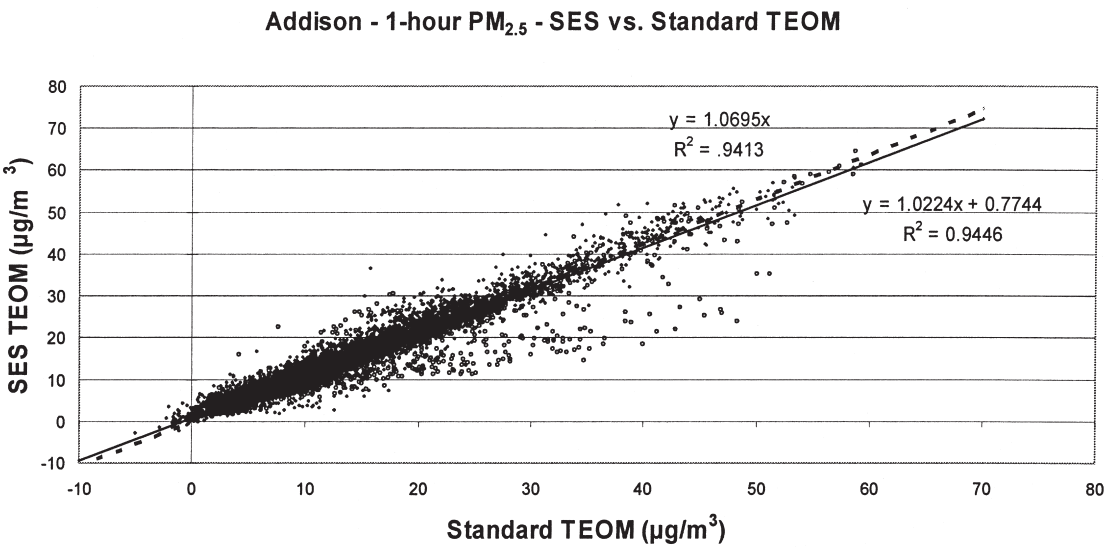


Figure 13. Correlation scatter plot of SES-equipped TEOM monitor vs. standard TEOM monitor hourly mass concentrations for Addison. Also shown are the unconstrained (solid) and zero-intercept (dashed) linear regression fits to the data, with the fitted equations and the R^2 values displayed on the figure.

swings in apparent mass concentration when the relative humidity changes rapidly. On balance, these correlations indicate that the SES-equipped TEOM instrument measures ~7% more mass than the standard TEOM instrument at this site, on average.

Figure 14 shows the results of regression fits to monthly scatter plots of SES-equipped versus standard TEOM monitor data for Addison. (Only the zero-intercept slopes of these regression lines are plotted—the full regression fits show a very similar pattern.) As noted, the SES-equipped TEOM monitor measures less mass than the standard TEOM monitor during the months of May and June 2001. There is a weak but discernable seasonal pattern in the slopes shown in Figure 14, with the largest slopes occurring during the cold winter months.

Figure 15 shows the scatter plot of all hourly TEOM monitor data for Queens, and regression slopes from zero-intercept monthly scatter plots of SES-equipped TEOM versus standard TEOM monitor data for Queens are presented in Figure 16. As with Figure 13, removal of the data from mid-April through July of 2001 yields a tighter correlation and slopes ~5% larger than illustrated in Figure 15. The correlation in Figure 15 indicates that the SES-equipped TEOM instrument measures ~8% more mass than the standard TEOM instrument at this site, on average. It is more difficult to assess the seasonal pattern at this site for this comparison. It appears that more data

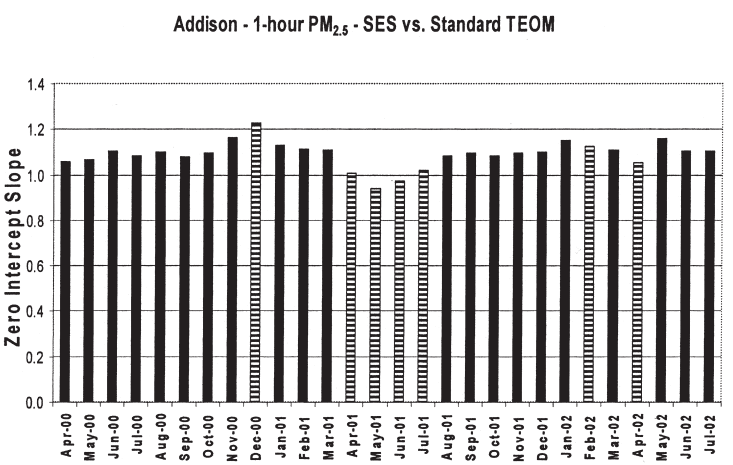


Figure 14. Zero-intercept regression slopes for the SES-equipped vs. standard TEOM monitor hourly data at Addison broken down by month. Dark bars indicate fits with an R^2 greater than 0.9, and striped bars indicate an R^2 between 0.8 and 0.9.

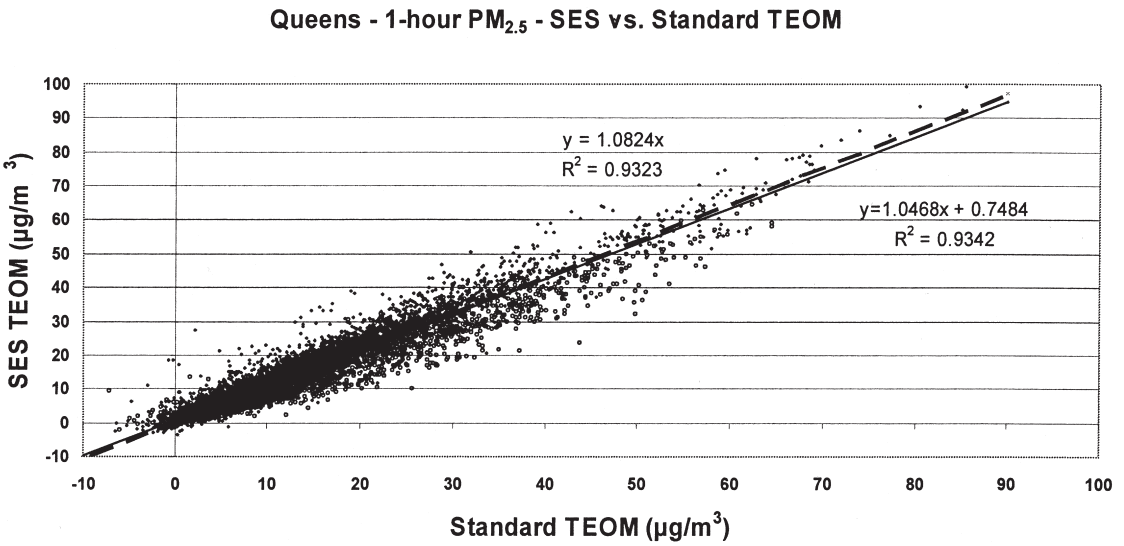


Figure 15. Correlation scatter plot of SES-equipped vs. standard TEOM monitor hourly mass concentrations for Queens. Also shown are the unconstrained (solid) and zero-intercept (dashed) linear regression fits to the data, with the fitted equations and the R^2 values displayed on the figure.

would be required to confirm or reject the notion of a consistent seasonal difference between the two TEOM instruments at this site.

The obvious features in Figure 16 are the slopes that are significantly below 1 for May and June 2001. As noted, this result is unexpected and requires some explanation. Possible explanations include (1) systematic error of one or both TEOM monitors, and (2) unusual composition of aerosol (or gases), which contribute to a systematic error evident only during these two months. Explanation 1 is unlikely, because it would require two sets of instruments at locations hundreds of kilometers apart to change systematically in concert. To further investigate explanation 2, future work will examine the ion balance of speciation filter measurements for that period to estimate the acidity/alkalinity of the aerosol. Nafion can act as an ion exchange resin and, as such, can potentially interact with aerosol. Some types of aerosol could possibly lose their particle-bound water more completely to the Nafion dryer than to the 50 °C heated and ventilated filter. Investigation of the chemical composition of the aerosol during that time period may help explain the unexpected results observed during this short time period. Experiential evidence indicates that selective reactivity of the Nafion is generally not a problem, but it cannot be ruled out in this case.

DISCUSSION AND SUMMARY

Measurements using the TEOM mass monitors are highly correlated with the FRM filter measurements for both the

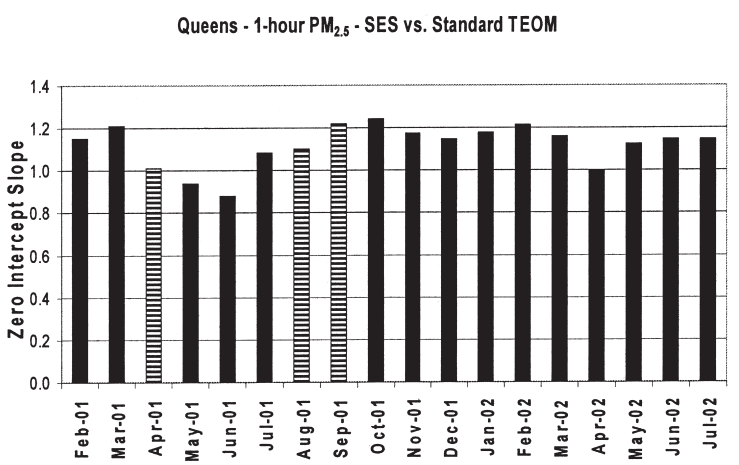


Figure 16. Zero-intercept regression slopes for the SES-equipped vs. standard TEOM monitor hourly data at Queens broken down by month. Dark bars indicate fits with an R^2 greater than 0.9, and striped bars indicate an R^2 between 0.8 and 0.9.

rural and the urban sites used in this study. Location-and season-dependent investigations of these correlations have been performed. The rural site in Addison shows a consistent negative bias of the TEOM monitor measurements with respect to FRM filter measurements. Using bias estimates that combine the regression results and the monthly average $PM_{2.5}$ mass concentration from FRM filters, the measurements using the TEOM monitor are biased low compared with the manual filter samples for all 34 months studied. It is hypothesized that the winter bias is caused in large part by volatilization of NH_4NO_3 from the TEOM sensor head and that the summer bias is caused by volatilization of photochemically produced organic species. The smallest estimated biases at this site are during the spring and fall transition months.

The urban site in Queens offers a more complicated picture. While the average bias for the 18-month period is quite similar to the rural site, the seasonal differences are more dramatic. The bias is strongly negative during the winter months. During the summer, the bias is more likely to be positive than negative for the SES-equipped TEOM monitor, which seems to indicate that this site has a different mix of semi-volatile compounds than the rural site. This could be because of traffic-related particles, which are not highly volatile after they are aged in the atmosphere for minutes to hours. The TEOM mass monitor and the FRM agree quite well during the summer months in Queens.

The SES-equipped TEOM instrument captures ~7–11% more mass than the standard TEOM instrument operated at 50 °C, depending on how the averaging is done. The additional mass measured by the SES-equipped TEOM monitor does not always correlate well with the mass measured by the standard 50 °C TEOM monitor, particularly on short time scales of less than a day. The volatility of aerosols varies widely depending on the sources, temperature, and age of the aerosol. Meteorological inversions can result in high concentrations of volatile primary emission pollutants trapped close to the ground. In these cases, the SES-equipped TEOM monitor measurements may be 15 $\mu g/m^3$ or more higher than the standard TEOM monitor measurements. This short-term discrepancy between the two types of TEOM monitors can be critically important when agencies are producing health advisories such as the air quality index. Many of the points above and to the left of the regression lines in Figures 13 and 15 are examples of when this additional mass may be enough to change the air quality index from good to moderate or even from moderate to unhealthy.

As evidenced by other studies,[12,14,16] the types of comparisons presented in this work depend strongly on location and the composition of the ambient aerosol typical of the site and region. Extrapolating the results of this study too widely will certainly lead to erroneous conclusions. Still, the results presented here are self-consistent and consistent with other published data for the northeast United States.[14] For these reasons, the basic conclusions are likely to be representative of much of the northeast United States.

Finally, this work also further emphasizes the need for a $PM_{2.5}$ measurement technique that is free of artifacts and interferences and can be widely deployed as part of state and local networks.[31,32] The SES-equipped TEOM monitor configuration evaluated in this work improves upon the standard TEOM monitor but still suffers from artifact and interferences similar to those experienced by the FRM filter samplers.

ACKNOWLEDGMENTS

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REFERENCES

1. Jayne, J.T.; Leard, D.C.; Zhang, X.; Davidovits, P.; Smith, K.A.; Kolb, C.E.; Worsnop, D.R. Development of an Aerosol Mass Spectrometer for Size and Composition Analysis of Submicron Particles; *Aerosol Sci. Technol.* **2000**, *33*, 49-70.
2. Weber, R.J.; Orsini, D.; Daun, Y.; Lee, Y.-N.; Klotz, P.J.; Brechtel, F.A. Particle-into-Liquid Collector for Rapid Measurement of Aerosol Bulk Chemical Composition; *Aerosol Sci. Technol.* **2001**, *35*, 718-727.
3. Stolzenburg, M.R.; Hering, S.V.A. New Method for the Automated Measurement of Atmospheric Fine Particle Nitrate; *Environ. Sci. Technol.* **2000**, *34*, 907-914.
4. Drewnick, F.; Schwab, J.J.; Hogrefe, O.V.; Peters, S.; Weber, R.; Diamond, D.; Husain, L.; Demerjian, K.L. Intercomparison and Evaluation of Four Semi-Continuous $PM_{2.5}$ Sulfate Instruments; *Atmos. Environ.* **2003**, *37*, 3335-3350.
5. Solomon, P.; Baumann, K.; Edgerton, E.; Tanner, R.; Eatough, D.; Modey, W.; Marin, H.; Savoie, D.; Natarajan, S.; Meyer, M.B.; Norris, G. Comparison of Integrated Samplers for Mass and Composition during the 1999 Atlanta Supersites Program; *J. Geophys. Res.* **2003**, *108*, 8423-8448.
6. 40 CFR Part 50; National Ambient Air Quality Standards for Particulate Matter; Final Rule; *Fed. Regist.* **1997**, *62* (138), 2-102.
7. Understanding the Health Effects of Components of the Particulate Matter Mix: Progress and Next Steps. In *HEI Perspectives;* Health Effects Institute: Cambridge, MA, 2002; pp 1-20.
8. Dockery, D.W.; Pope, C.A., III. Acute Respiratory Effects of Particulate Air Pollution; *Annu. Rev. Public Health* **1994**, *15*, 107-132.
9. Schwartz, J. What Are People Dying of on High Air Pollution Days?; *Environ. Res.* **1994**, *64*, 26-35.
10. Katsouyanni, K.; Karakatsani, A.; Messari, I.; Touloumi, G.; Hatzakis, A.; Kalandidi, A.; Trichopoulos, D. Air Pollution and Cause-Specific Mortality in Athens; *J. Epidemiol. Commun. Health* **1990**, *44*, 321-324.

11. 40 CFR Parts 53 and 58; Revised Requirements for Designation of Reference and Equivalent Methods for $PM_{2.5}$ and Ambient Air Quality Surveillance for Particulate Matter; Final Rule; *Fed. Regist.* **1997**, *62* (138), 2-102.
12. Pang, Y.; Ren, R.; Obeidi, F.; Hastings, R.; Eatough, D.J.; Wilson, W. Semi-Volatile Species in $PM_{2.5}$: Comparison of Integrated and Continuous Samplers for $PM_{2.5}$ Research or Monitoring; *J. Air & Waste Manage. Assoc.* **2001**, *51*, 25-36.
13. Pang, Y.; Eatough, N.L.; Eatough, D.J. $PM_{2.5}$ Semivolatile Organic Material at Riverside, California: Implications for the $PM_{2.5}$ Federal Reference Method Sampler; *Aerosol Sci. Technol.* **2002**, *36*, 277-288.
14. Allen, G.; Sioutas, C.; Koutrakis, P.; Reiss, R. Lurmann, F.W.; Roberts, P.W. Evaluation of the TEOM Method for Measurement of Ambient Particulate Mass in Urban Areas; *J. Air & Waste Manage. Assoc.* **1997**, *47*, 682-689.
15. Patashnick, H. Letters to the Editor; *J. Air & Waste Manage. Assoc.* **1998**, *48*, 195-200.
16. Chung, A.; Chang, D.P.Y.; Kleeman, M.J.; Perry, K.D.; Cahill, T.A.; Dutcher, D.; McDougall, E.M.; Stroud, K. Comparison of Real-Time Instruments Used to Monitor Airborne Particulate Matter; *J. Air & Waste Manage. Assoc.* **2001**, *51*, 109-120.
17. Mignacca, D.; Stubbs, K. Effects of Equilibration Temperature on PM_{10} Concentrations from the TEOM Method in the Lower Fraser Valley; *J. Air & Waste Manage. Assoc.* **1999**, *49*, 1250-1254.
18. U.S. Environmental Protection Agency. Monitoring $PM_{2.5}$ in Ambient Air Using Designated Reference or Class I Equivalent Methods. In *Quality Assurance Handbook for Air Pollution Measurement Systems;* U.S. Government Printing Office: Washington, DC, 1998; Volume II, Part 2, Section 2.12.
19. Noble, C.A.; Vanderpool, R.W.; Peters, T.M.; McElroy, F.F.; Gemmill, D.B.; Wiener, R.W. Federal Reference and Equivalent Methods for Measuring Fine Particulate Matter; *Aerosol Sci. Technol.* **2001**, *34*, 457-464.
20. Peters, T.M.; Norris, G.A.; Vanderpool, R.W.; Gemmill, D.B.; Wiener, R.W.; Murdoch, R.W.; McElroy, F.F.; Pitchford, M. Field Performance of $PM_{2.5}$ Federal Reference Method Samplers; *Aerosol Sci. Technol.* **2001**, *34*, 433-443.
21. Patashnick, H.; Rupprecht, E.G. Continuous PM_{10} Measurements Using the Tapered Element Oscillating Microbalance; *J. Air & Waste Manage. Assoc.* **1991**, *41*, 1079-1083.
22. *Methods of Air Sampling and Analysis*; 4th ed.; Lodge, J.P., Jr., Watson, J., Chow, J., Eatough, D., Eds.; submitted for publication. Lewis Publishers, CRC Press: Boca Raton, FL.
23. Meyer, M.B.; Patashnick, H.; Ambs, J.L.; Rupprecht, E. Development of a Sample Equilibration System for the TEOM Continuous PM Monitor; *J. Air & Waste Manage. Assoc.* **2000**, *50*, 1345-1349.
24. Kim, B.M.; Teffera, S.; Zeldin, M.D. Characterization of $PM_{2.5}$ and PM_{10} in the South Coast Air Basin of Southern California: Part 2-Temporal Variations; *J. Air & Waste Manage. Assoc.* **2000**, *50*, 2045-2059.
25. Sheridan, P.J.; Delene, D.J.; Ogren, J.A. Four Years of Continuous Surface Aerosol Measurements for the DOE/ARM Southern Great Plains CART Site; *J. Geophys. Res.* **2001**, *106D*, 20,735-20,747.
26. Quinn, P.K.; Miller, T.L.; Bates, T.S.; Ogren, J.A.; Andrews, E.; Shaw, G.E. A 3-Year Record of Simultaneously Measured Aerosol Chemical and Optical Properties at Barrow, Alaska; *J. Geophys. Res.* **2002**, *107D*, 4130.
27. Schwab, J.J.; Felton, H.D.; Demerjian, K.L. Aerosol Chemical Composition in New York State from Integrated Filter Samples: Urban/Rural and Seasonal Contrasts; *J. Geophys. Res.,* in press, DOI 10.1029/2003JD004078.
28. Tanner, R.L.; Parkhurst, W.J. Chemical Composition of Fine Particles in the Tennessee Valley Region; *J. Air & Waste Manage. Assoc.* **2000**, *50*, 1299-1307.
29. Van Loy, M.; Bahadori, T.; Wyzga, R.; Hartsell, B.; Edgerton, E. The Aerosol Research and Inhalation Epidemiology Study (ARIES): $PM_{2.5}$ Mass and Aerosol Component Concentrations and Sampler Intercomparisons; *J. Air & Waste Manage. Assoc.* **2000**, *50*, 1446-1458.
30. Schwab, J.J.; Ambs, J.L.; Hogrefe, O.; Demerjian, K.L. Laboratory Characterization of Modified TEOM Samplers; *J. Air & Waste Manage. Assoc.*, **2004**, *54*, 1254-1263.
31. Eatough, D.J.; Eatough, N.L.; Obeidi, F.; Pang, Y.; Modey, W.; Long, R. Continuous Determination of $PM_{2.5}$ Mass, Including Semi-Volatile Species; *Aerosol Sci. Technol.* **2001**, *34*, 1-8.
32. Patashnick, H.; Rupprecht, G.; Ambs, J.L.; Meyer, M.B. Development of a Reference Standard for Particulate Matter Mass in Ambient Air; *Aerosol Sci. Technol.* **2001**, *34*, 42-45.

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